

CLASSIFICATION OF INTER-ICTAL AND ICTAL EEG SIGNALS USING MAXIMUM OVERLAP DISCRETE WAVELET TRANSFORM AND ANN CLASSIFIER

a project report submitted by

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled, “**CLASSIFICATION OF INTER-ICTAL AND ICTAL EEG SIGNALS USING MAXIMUM OVERLAP DISCRETE WAVELET TRANSFORM AND ANN CLASSIFIER** ” is a bonafide record of work of the following candidates who carried out the project work under my supervision during the academic year 2019-2020.

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ABSTRACT

Background: Epilepsy is the second most common neurological disorder affecting approximately 60 million people worldwide. Seizure is a neurological disorder of the brain and this occurs when the human brain cell fires electrical impulses three times more than that of a normal one, a pattern of repeated seizure is called epilepsy. Epilepsy can be caused by alcohol, drugs, brain injury and can also be genetically carried. Electroencephalography (EEG) is the easiest, cheapest, and most effective practice for providing information related to epilepsy diagnosis.

Method: To conduct our experiments, we utilize university of Bonn's widely used and publicly available epilepsy data-set in this work. The data-set contains 500 single-channel EEG signal segments. Each of the segment is 23.6-s in duration. The recordings have been grouped into five sets, namely A, B, C, D, and E, each of which contains 100 EEG epochs. We have developed a method using LBP, LGP pattern techniques and Kruskal-Wallis test to extract the best features from the EEG signal database and use it to train an ANN classifier based on Conjugate Gradient Algorithm which is a Backpropagation algorithm proposed by Powell, with an objective to obtain the highest accuracy during testing.

Result: Experimental outcomes suggest that the algorithmic performance of the proposed scheme is promising in all the cases of clinical significance. Comparative evaluation of algorithmic performance with the state-of-the-art schemes manifest that the seizure detection scheme proposed herein outperforms competing algorithms in terms of accuracy and sensitivity.

Conclusion: Upon the implementation of the proposed method, it can be used to classify the different stages of epilepsy and can also be used for other biomedical applications.

CHAPTER 1

INTRODUCTION

1.1 NEURON

A neuron is a nerve cell that is the basic building block of the nervous system. Neurons are similar to other cells in the human body in a number of ways, but there is one key difference between neurons and other cells. Neurons are specialized to transmit information throughout the body. These highly specialized nerve cells are responsible for communicating information in both chemical and electrical forms. There are also several different types of neurons responsible for different tasks in the human body.

A useful analogy is to think of a neuron as a tree. A neuron has three main parts: dendrites, an axon, and a cell body or soma (see image below), which can be represented as the branches, roots and trunk of a tree, respectively. A dendrite (tree branch) is where a neuron receives input from other cells. Dendrites branch as they move towards their tips, just like tree branches do, and they even have leaf-like structures on them called spines.

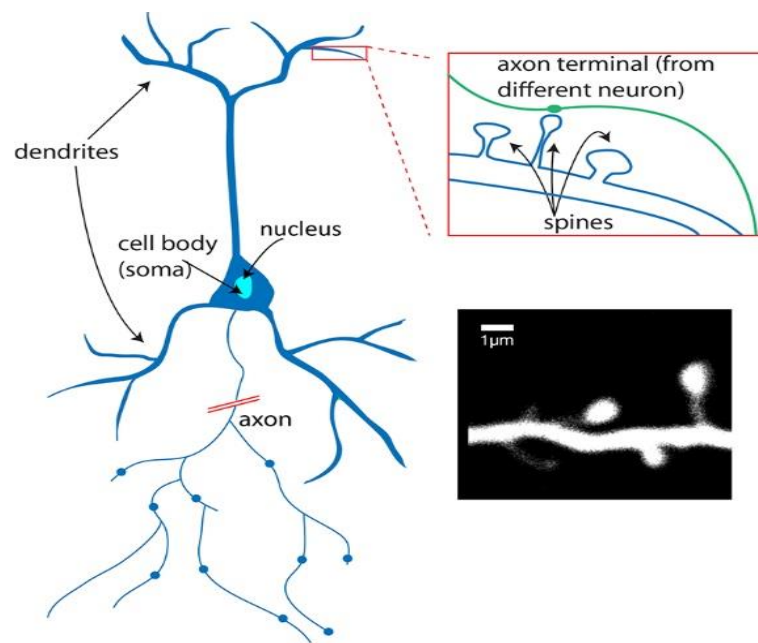


Figure 1

The axon (tree roots) is the output structure of the neuron; when a neuron wants to talk to another neuron, it sends an electrical message called an action potential throughout the entire axon. The soma (tree trunk) is where the nucleus lies, where the neuron's DNA is housed, and where proteins are made to be transported throughout the axon and dendrites.

The tree-like structure of a neuron. Dendritic spines are small structures that receive inputs from the axons of other neurons. Bottom-right image: a segment of dendrite from which spines branch off, like leaves off a tree branch. Note the very small size ($\sim 0.001\text{mm}$). Axons are the long, thin structure in which action potentials are generated; the transmitting part of the neuron. After initiation, action potentials travel down axons to cause release of neurotransmitter. The receiving part of the neuron. Dendrites receive synaptic inputs from axons, with the sum total of dendritic inputs determining whether the neuron will fire an action potential. Dendrites are known as Dendrites. Spines are the small protrusions found on dendrites that are, for many synapses, the postsynaptic contact site.

In order for neurons to communicate, they need to transmit information both within the neuron and from one neuron to the next. This process utilizes both electrical signals as well as chemical messengers. The dendrites of neurons receive information from sensory receptors or other neurons. This information is then passed down to the cell body and on to the axon. Once the information has arrived at the axon, it travels down the length of the axon in the form of an electrical signal known as an action potential (electrical impulses).

Normally, the inside of the cell is more negative than the outside; neuroscientists say that the inside is around -70 mV with respect to the outside, or that the cell's resting membrane potential is -70 mV . This membrane potential isn't static. It's constantly going up and down, depending mostly on the inputs coming from the axons of other neurons. Some inputs make the neuron's membrane potential become more positive (or less negative, e.g. from -70 mV to -65 mV), and others do the opposite. These are respectively termed excitatory and inhibitory inputs, as they promote or inhibit the generation of action potentials (the reason some inputs are excitatory and others inhibitory is that different types of neuron release different neurotransmitters; the neurotransmitter used by a neuron

determines its effect). Action potentials are the fundamental units of communication between neurons and occur when the sum total of all of the excitatory and inhibitory inputs makes the neuron's membrane potential reach around -50 mV (see diagram), a value called the action potential threshold. Neuroscientists often refer to action potentials as 'spikes', or say a neuron has 'fired a spike' or 'spiked'. The term is a reference to the shape of an action potential as recorded using sensitive electrical equipment.

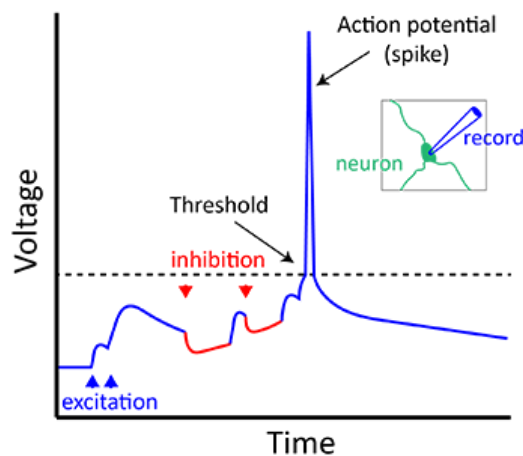


Figure 2

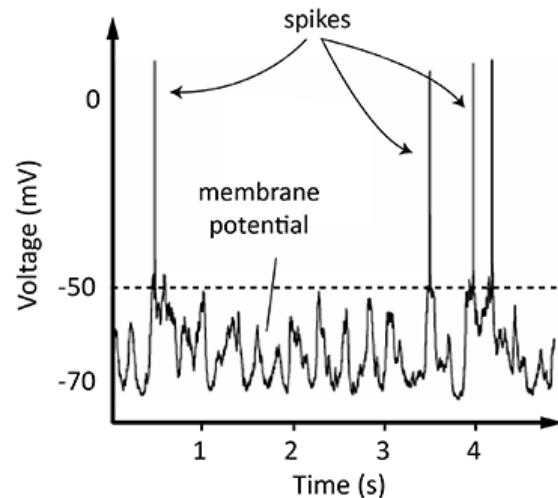


Figure 3

1.2 EEG

An EEG is a test that detects abnormalities in the brain waves, or in the electrical activity of brain. During the procedure, electrodes consisting of small metal discs with thin wires are pasted onto scalp. The electrodes detect tiny electrical charges that result from the activity of brain cells. The charges are amplified and appear as a graph on a computer screen, or as a recording that may be printed out on paper. The EEG is used to evaluate several types of brain disorders. When epilepsy is present, seizure activity will appear as rapid spiking waves on the EEG. People with lesions of their brain, which can result from tumors or stroke, may have unusually slow EEG waves, depending on the size and the location of the lesion. The test can also be used to diagnose other disorders that influence

brain activity, such as Alzheimer's disease, certain psychoses, and a sleep disorder called narcolepsy. The EEG may also be used to determine the overall electrical activity of the brain (for example, to evaluate trauma, drug intoxication, or extent of brain damage in comatose patients). The EEG may also be used to monitor blood flow in the brain during surgical procedures.

1.3 BRAIN WAVES

The brain has billions of neurons, and each individual neuron connects (on average) to thousands of others. Communication happens between them through small electrical currents that travel along the neurons and throughout enormous networks of brain circuits. When all these neurons are activated they produce electrical pulses visualize a wave rippling through the crowd at a sports arena this synchronized electrical activity results in a “brainwave”.

At the root of all our thoughts, emotions and behaviours is the communication between neurons within our brains. Brainwaves are produced by synchronised electrical pulses from masses of neurons communicating with each other. Our brainwaves change according to what we're doing and feeling. When slower brainwaves are dominant we can feel tired, slow, sluggish, or dreamy. The higher frequencies are dominant when we feel wired, or hyper-alert. Brainwave speed is measured in Hertz (cycles per second) and they are divided into bands delineating slow, moderate, and fast waves.

1.4 TYPES OF BRAIN WAVES

1.4.1 Infra-Low (<.5HZ)

Infra-Low brainwaves (also known as Slow Cortical Potentials), are thought to be the basic cortical rhythms that underlie our higher brain functions. Very little is known

about infra-low brainwaves. Their slow nature make them difficult to detect and accurately measure, so few studies have been done. They appear to take a major role in brain timing and network function.

1.4.2 Delta (.5 TO 3 HZ)

Delta brainwaves are slow, loud brainwaves (low frequency and deeply penetrating, like a drum beat). They are generated in deepest meditation and dreamless sleep. Delta waves suspend external awareness and are the source of empathy. Healing and regeneration are stimulated in this state, and that is why deep restorative sleep is so essential to the healing process.



1.4.3 Theta (3 TO 8 HZ)

Theta brainwaves occur most often in sleep but are also dominant in deep meditation. Theta is our gateway to learning, memory, and intuition. In theta, our senses are withdrawn from the external world and focused on signals originating from within. It is that twilight state which we normally only experience fleetingly as we wake or drift off to sleep. In theta we are in a dream; vivid imagery, intuition and information beyond our normal conscious awareness. It's where we hold our 'stuff', our fears, troubled history, and nightmares.



1.4.4 Alpha (8 TO 12 HZ)

Alpha brainwaves are dominant during quietly flowing thoughts, and in some meditative states. Alpha is ‘the power of now’, being here, in the present. Alpha is the resting state for the brain. Alpha waves aid overall mental coordination, calmness, alertness, mind/body integration and learning.



1.4.5 Beta (12 TO 38 HZ)

Beta brainwaves dominate our normal waking state of consciousness when attention is directed towards cognitive tasks and the outside world. Beta is a ‘fast’ activity, present when we are alert, attentive, engaged in problem solving, judgment, decision making, or focused mental activity.



Beta brainwaves are further divided into three bands; Lo-Beta (Beta1, 12-15Hz) can be thought of as a 'fast idle', or musing. Beta (Beta2, 15-22Hz) is high engagement or actively figuring something out. Hi-Beta (Beta3, 22-38Hz) is highly complex thought, integrating new experiences, high anxiety, or excitement. Continual high frequency processing is not a very efficient way to run the brain, as it takes a tremendous amount of energy.

1.4.6 Gamma (38 TO 42 HZ)

Gamma brainwaves are the fastest of brain waves (high frequency, like a flute), and relate to simultaneous processing of information from different brain areas. Gamma brainwaves pass information rapidly and quietly. The most subtle of the brainwave frequencies, the mind has to be quiet to access gamma.



Gamma was dismissed as 'spare brain noise' until researchers discovered it was highly active when in states of universal love, altruism, and the 'higher virtues'. Gamma is also above the frequency of neuronal firing, so how it is generated remains a mystery. It is speculated that gamma rhythms modulate perception and consciousness, and that a greater presence of gamma relates to expanded consciousness and spiritual emergence.

1.5 EPILEPSY

Epilepsy is a central nervous system (neurological) disorder in which brain activity becomes abnormal, causing seizures or periods of unusual behavior, sensations, and sometimes loss of awareness. Anyone can develop epilepsy. Epilepsy affects both males and females of all races, ethnic backgrounds and ages. Seizure symptoms can vary widely. Some people with epilepsy simply stare blankly for a few seconds during a seizure, while others repeatedly twitch their arms or legs. Having a single seizure doesn't mean you have epilepsy. At least two unprovoked seizures are generally required for an epilepsy diagnosis. Treatment with medications or sometimes surgery can control seizures for the majority of people with epilepsy. Some people require lifelong treatment to control seizures, but for

others, the seizures eventually go away. Some children with epilepsy may outgrow the condition with age.

1.5.1 TYPES OF SEIZURES

Seizures are mainly divided into two types namely generalized seizures and Focal (or partial) seizures. Generalized seizures affect both sides of the brain. Absence seizures, sometimes called petit mal seizures, can cause rapid blinking or a few seconds of staring into space. Tonic-clonic seizures, also called grand mal seizures, can make a person Cry out, lose consciousness, fall to the ground, have muscle jerks or spasms, the person may feel tired after a tonic-clonic seizure. This generalized seizure is the most common type that is seen in people. This type of seizure is curable with some treatment and medications. Focal seizures are located in just one area of the brain. These seizures are also called partial seizures. Simple focal seizures affect a small part of the brain. These seizures can cause twitching or a change in sensation, such as a strange taste or smell. Complex focal seizures can make a person with epilepsy confused or dazed. The person will be unable to respond to questions or direction for up to a few minutes. Secondary generalized seizures begin in one part of the brain, but then spread to both sides of the brain. In other words, the person first has a focal seizure, followed by a generalized seizure. This Seizures may last as long as a few minutes. Focal seizures cannot be cured, its incurable. Sometimes it can be cured by doing the open brain surgery.

1.5.2 INTERICTAL

The abnormal EEG activity that occurs between seizures is known as Interictal period. Epileptic form activity includes spikes, spike and waves, and polyspike complexes. Abnormal sharp waves are also considered epileptic form or Interictal activity. The EEG signals are obtained through a number of electrodes that are placed on the scalp of an epileptic patient and are recorded during the interval of two seizures. There are two sets of signals recorded, one taken from the focal region of the seizure, the other taken from the opposite hemisphere of the focal region. Both the signals are used in the classification of Ictal and Interictal signals.

1.5.3. ICTAL

Ictal psychiatric symptoms are most commonly associated with partial seizures, although they can also occur with generalized seizures. Anxiety, fear, and psychosis are the most common psychiatric symptoms experienced during a seizure. Anxiety or fear is a component of the aura in one third of patients with partial seizures; the anxiety is often intense and may last throughout the course of the seizure. It may be most common in patients with right temporal foci.

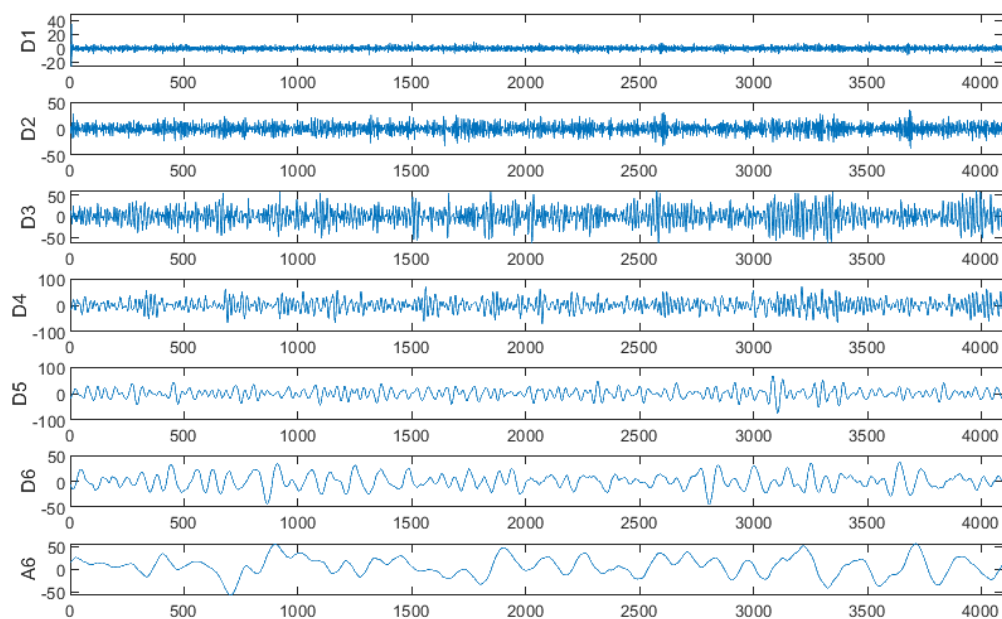


Figure 4

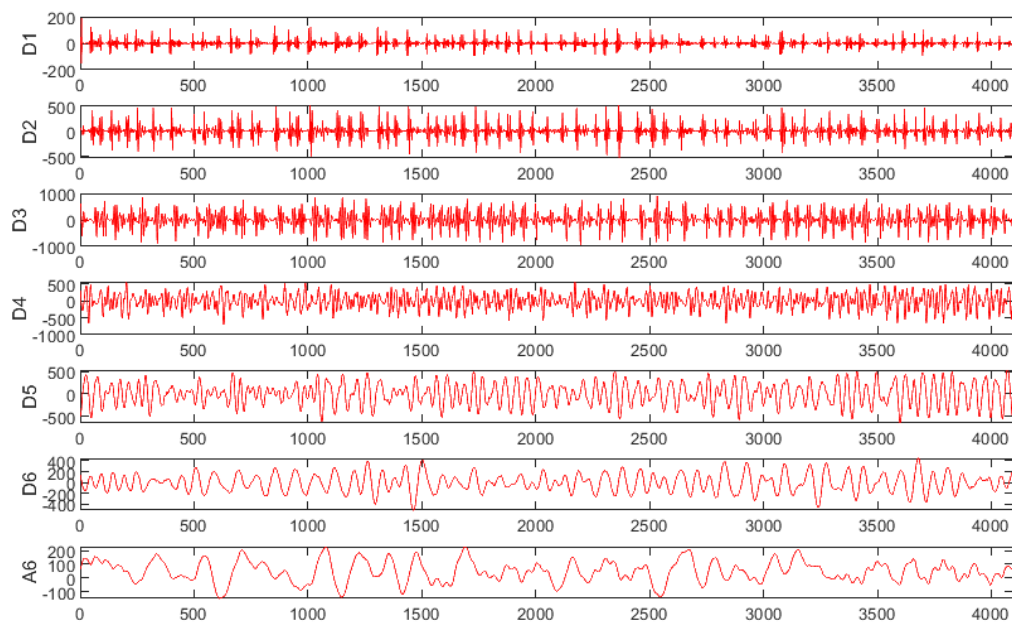


Figure 5

CHAPTER 2

LITERATURE SURVEY

Researches on automatic epileptic seizure detection systems date back to 1970s. Various studies have been done until now. Costa et al. (Costa et al., 2008) tried to classify patients into four possible epileptic behaviors named inter-ictal, pre-ictal, ictal and postictal. In order to achieve successful results, they have utilized EEG signals and extracted only 14 features from different concepts. Of these features, four were extracted with signal energy, eight extracted using wavelet transform and last two attained from using nonlinear system dynamics. After normalizing, these features were fed to 6 different types of artificial neural network (ANN) with 3 different success metrics named sensitivity, specificity and accuracy. Two different patient data sets were used for training and tests. Authors concluded that 99% of success achieved. However, their classifier developed for one patient has failed in the test of other patients, presenting patient-specific classifier.

This paper (San-Segundo et al., 2019) describes the analysis of a deep neural network for the classification of epileptic EEG signals. The deep learning architecture is made up of two convolutional layers for feature extraction and three fully-connected layers for classification. We evaluated several EEG signal transforms for generating the inputs to the deep neural network: Fourier, wavelet and empirical mode decomposition. This analysis was carried out using two public datasets (Bern-Barcelona EEG and Epileptic Seizure Recognition datasets) obtaining significant improvements in accuracy. For the Bern-Barcelona EEG, we obtained an increase in accuracy from 92.3% to 98.9% when classifying between focal and non-focal signals using the empirical mode decomposition. For the Epileptic Seizure Recognition dataset, we evaluated several scenarios for seizure detection obtaining the best results when using the Fourier transform. The accuracy increased from 99.0% to 99.5% for classifying non-seizure vs. seizure recordings, from 91.7% to 96.5% when differentiating between healthy, non-focal and seizure recordings, and from 89.0% to 95.7% when considering healthy, focal and seizure recordings.

In the work of Tzallas et al.(Tzallas et al., 2007), it has been aimed to detect existence of the seizure. In order to attain better results, first of all, time frequency (TF) analysis of the EEGs was done to get the spectrums. Features of these spectrums were extracted using fractional energy on TF windows. Smoothed pseudo-Wigner–Ville distribution (SPWVD) was exploited to draw out features from spectrums. Obtained features were further fed to principal component analysis (PCA) to reduce the dimensions of the features. Afterwards, feed-forward ANN was utilized to classify the signals. Performance metrics such as sensitivity, specificity and selectivity were used and results ranging from 97% to 100% were attained. Besides, some artifacts in the EEGs have been removed manually after visual inspection. Authors asserted that their methodology has promising potential and needs further evaluation.

In another study(Jaiswal & Banka, 2017), Jaiswal and Banka introduced two feature extraction techniques based on local pattern transformation for classification of epileptic EEG signals, namely one dimensional local gradient pattern (1D-LGP) and local neighbor descriptive pattern (LNDP). Classification performances of the proposed techniques were evaluated in a 10-fold cross validation scheme using nearest neighbor, SVM, decision tree and ANN. Their results shows that LNDP and 1D-LGP feature extraction techniques together with ANN classifier attained the average classification accuracies of 99.82% and 99.80%, respectively They introduced two novel feature extraction techniques in another study. Local centroid pattern (LCP) and one-dimensional local ternary pattern (1D-LTP) techniques were proposed to overcome 1D-LBP technique's sensitivity to local variation. Their histogram-based feature vectors were reportedly simple and easy to implement. Their proposed methods have been tested on nine different experimental cases to validate their effectiveness. Their classification accuracies are also reported to be ranging from 97.5% to 100% for these cases.

In the study presented by Murugavel and Ramakrishnan (Murugavel & Ramakrishnan, 2011), the same dataset w(Bonn University) was employed to classify epileptic seizure making use of wavelet transform, approximate entropy (ApEn) and

probabilistic neural network (PNN). After exposing EEG signals to wavelet transform, their features were extracted using ApEn, and the best thirty features were selected for further to be processed in ANNs. This new selection process involves choosing the features having the minimal variance within the class and maximum absolute difference between classes. System evaluation metric was only based on the accuracy as high as 100%.

Guo et al. (Guo et al., 2010) explored a new scheme of automated seizure detection for epilepsy disease. They classified the database as the three cases. The scheme utilized wavelet transform in the feature extraction phase producing line length features, and ANN to classify EEGs. Firstly, the EEG signals were decomposed into several sub-bands through discrete wavelet transform (DWT). Afterwards, line length features were drawn out from these sub-signals. Finally, multilayer perceptron neural network (MLPNN) with backpropagation training algorithm was employed to classify EEGs. The classification accuracy of 99.6% was achieved using that method.

In this study,(Ocak, 2009) a new scheme was presented for detecting epileptic seizures from electro-Encephalogram (EEG) data recorded from normal subjects and epileptic patients. The new scheme was based on approximate entropy (ApEn) and discrete wavelet transform (DWT) analysis of EEG signals. Seizure detection was accomplished in two stages. In the first stage, EEG signals were decomposed into approximation and detail coefficients using DWT. In the second stage, ApEn values of the approximation and detail coefficients were computed. Significant differences were found between the ApEn values of the epileptic and the normal EEG allowing us to detect seizures with over 96% accuracy. Without DWT as preprocessing step, it was shown that the detection rate was reduced to 73%. The analysis results depicted that during seizure activity EEG had lower ApEn values compared to normal EEG. This suggested that epileptic EEG was more predictable or less complex than the normal EEG. The data was further analyzed with surrogate data analysis methods to test for evidence of nonlinearities. It was shown that epileptic EEG had significant nonlinearity whereas normal EEG behaved similar to Gaussian linear stochastic process.

This paper(Subasi, 2005) deals with a novel method of analysis of EEG signals using discrete wavelet transform, and classification using ANN. EEG signals were decomposed into the frequency sub-bands using wavelet transform. Then these sub-band frequencies were used as an input to an ANN with two discrete outputs: normal and epileptic. In this study, FEBANN and DWN based classifiers were developed and compared in relation to their accuracy in classification of EEG signals. The comparisons between the developed classifiers were primarily based on analysis of the ROC curves as well as a number of scalar performance measures pertaining to the classification. The DWN-based classifier outperformed the FEBANN based counterpart. Within the same group, the DWN-based classifier was more accurate than the FEBANN based classifier. The DWN-based classifier identified accurately all the epileptic and normal cases with specificity 93.1% and sensitivity 92.8% and the FEBANN-based classifier with specificity 91.3% and sensitivity 90.4%.

In this paper(Übeyli, 2008) a new approach based on the implementation of multiclass support vector machine (SVM) with the error correcting output codes (ECOC) is presented for classification of electroencephalogram (EEG) signals. In practical applications of pattern recognition, there are often diverse features extracted from raw data which needs recognizing. Decision making was performed in two stages: feature extraction by eigenvector methods and classification using the classifiers trained on the extracted features. The aim of the study is classification of the EEG signals by the combination of eigenvector methods and multiclass SVM. The purpose is to determine an optimum classification scheme for this problem and also to infer clues about the extracted features. This research demonstrated that the eigenvector methods are the features which well represent the EEG signals and the multiclass SVM trained on these features achieved high classification accuracies.

This paper(Ubeyli, 2005) describes the application of adaptive neuro-fuzzy inference system (ANFIS) model for classification of electroencephalogram (EEG) signals. Decision making was performed in two stages: feature extraction using the wavelet

transform (WT) and the ANFIS trained with the backpropagation gradient descent method in combination with the least squares method. Five types of EEG signals were used as input patterns of the five ANFIS classifiers. To improve diagnostic accuracy, the sixth ANFIS classifier (combining ANFIS) was trained using the outputs of the five ANFIS classifiers as input data. The proposed ANFIS model combined the neural network adaptive capabilities and the fuzzy logic qualitative approach. Some conclusions concerning the saliency of features on classification of the EEG signals were obtained through analysis of the ANFIS. The performance of the ANFIS model was evaluated in terms of training performance and classification accuracies and the results confirmed that the proposed ANFIS model has potential in classifying the EEG signals. S. The total classification accuracy they got for the ANFIS model was 98.68%.

In this work(Sadati et al., 2006), we proposed a neural fuzzy network for classification of epileptic and normal EEG signals. This is a neural network followed by an ANSIF as its activation function. After segmentation of EEG signals, they were decomposed into Time–frequency representations using DWT in five levels. The energy of details of five levels and the approximation of fifth level were used as the classifier inputs. SVM, FBNN, ANFIS and the proposed network (ANFIS) were implemented for the classification of EEG signals using DWT sub-bands as inputs. The aim of this work is to compare the different classifiers when applied to EEG data from normal and epileptic subjects. For this purpose an adaptive neural fuzzy network (ANFIS) to classify normal and epileptic EEG signals is proposed. The results are compared with other classifiers such as SVM (Support Vector Machine), ANFIS and FBNN (Feed forward Back-propagation Neural Network). It is shown that a classification accuracy of about 85.9% can be achieved using ANFN.

In this paper(Srinivasan et al., 2005) a recurrent type of neural network known as Elman network (EN) has been employed for the detection of epilepsy. Pre-processed EEG segments of 1 s duration have been used as test patterns. Five different features comprising two time-domain and three frequency-domain features are used in evaluating the

performance of the neural networks. Five types of training schemes have been employed for training the neural network. The overall detection accuracy obtained with EN using a single attribute is better than that obtained with the LAMSTAR network using two attributes. The experimental results show that the epileptic detection can be carried out with an accuracy rate as high as 99.6% even with a single input feature.

In this study (Zhou et al., 2018), a convolutional neural network (CNN) based on raw EEG signals instead of manual feature extraction was used to distinguish ictal, preictal, and interictal segments for epileptic seizure detection. They compared the performances of time and frequency domain signals in the detection of epileptic signals based on the intracranial Freiburg and scalp CHB-MIT databases to explore the potential of these parameters. Three types of experiments involving two binary classification problems (interictal vs. preictal and interictal vs. ictal) and one three-class problem (interictal vs. preictal vs. ictal) were conducted to explore the feasibility of this method. Using frequency domain signals in the Freiburg database, average accuracies of 96.7, 95.4, and 92.3% were obtained for the three experiments, while the average accuracies for detection in the CHB-MIT database were 95.6, 97.5, and 93% in the three experiments. Using time domain signals in the Freiburg database, the average accuracies were 91.1, 83.8, and 85.1% in the three experiments, while the signal detection accuracies in the CHB-MIT database were only 59.5, 62.3, and 47.9% in the three experiments. This method is obviously time-consuming and tedious, and a timely process.

The aim of this study (Polat & Gu, 2007) is to detect epileptic seizure in EEG signals using a hybrid system based on decision tree classifier and fast Fourier transform (FFT). The present study proposes a hybrid system with two stages: feature extraction using FFT and decision making using decision tree classifier. The detection of epileptic form discharges in the electroencephalogram (EEG) is an important part in the diagnosis of epilepsy. All data set were obtained from EEG signals of healthy subjects and subjects suffering from epilepsy diseases. For healthy subjects is background EEG (scalp) with open eyes and for epileptic patients correspond to a seizure recorded in hippocampus (epileptic

focus) with depth electrodes. The evolution of proposed system was conducted using k-fold cross-validation, classification accuracy, and sensitivity and specificity values. They have obtained 98.68% and 98.72% classification accuracies using 5- and 10-fold cross-validation. The stated results show that the proposed method could point out the ability of design of a new intelligent assistance diagnosis system.

In this paper(Li et al., 2017), an approach based on the maximal overlap discrete wavelet transform (MODWT) and log-normal distribution (LND) model was proposed for identifying epileptic seizures from electroencephalogram (EEG) signals. To carry out this study, they explored the potentials of MODWT to decompose the signals into time-frequency sub-bands till sixth level. And demodulation analysis (DA) was investigated to reveal the subtle envelope information from the sub-bands. All obtained coefficients are then used to calculate LND features, scale parameter (s) and shape parameter (m), which were fed to a random forest classifier (RFC) for classification. They have proposed an approach based on MODWT and LND model, which was successfully applied in automatic epilepsy detection. For various clinically relevant cases, such as A\E, AB\CDE, ABCD\E and AB\CD\E, 100%, 98.48%, 99.10%, and 98.10% are achieved with the RFC, respectively.

CHAPTER 3

METHODOLOGY

BASIC BLOCK OF THE CLASSIFICATION PROCESS

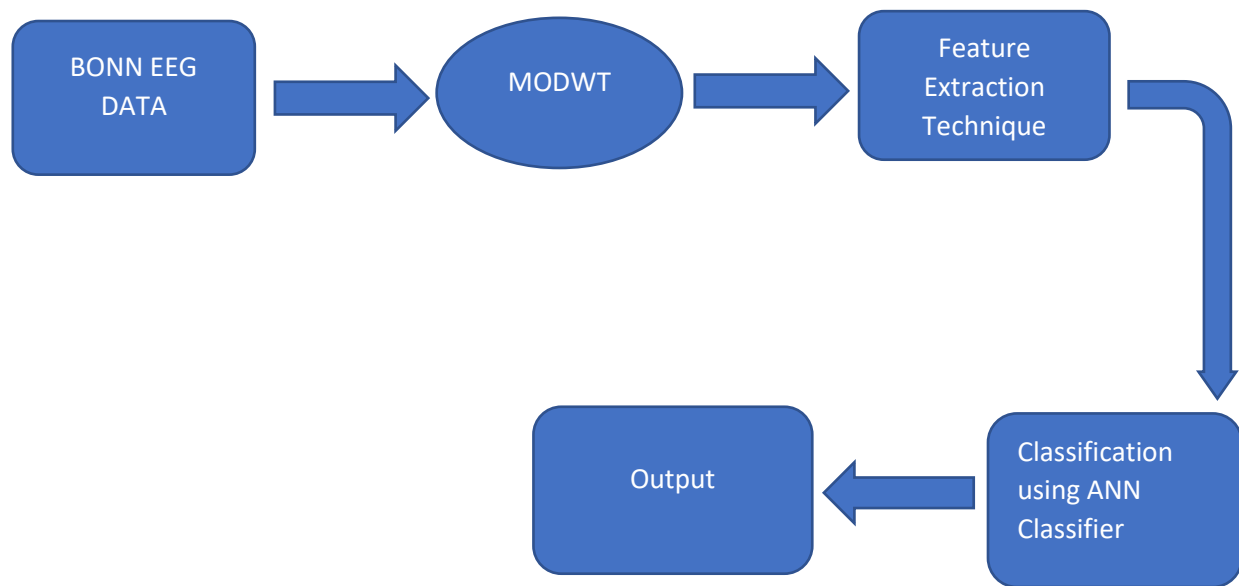


Figure 6

In general EEGs are divided into different frequency ranges or brain rhythms. In healthy adults itself, the amplitude and frequencies of such rhythms will change from one mental state to another, such as wakefulness to sleep or relaxed to active thinking. These frequency bands are designated as delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz) and beta (13–26 Hz). The “10–20” system is an internationally recognized method to describe the placing of scalp electrodes for recording EEG. The system is based on the relationship between the location of an electrode and the underlying area of cerebral cortex.

3.1 EEG Database :

The Epilepsy Center of the Bonn University Hospital of Freiburg [29] has provided a benchmark EEG database which is employed to determine the feasibility of our proposed technology. This database contains five subsets and they are named as Sets A–E, respectively. Each subset consists of 100 isometric single-channel segments. The duration of each segment is 23.6 s, the sampling frequency is 173.61 Hz, and the length of each record is $23.6 \times 173.61 \approx 4097$. EEGs of the first two subsets (Sets A and B) were captured extra-cranially using the international 10–20 system from five healthy volunteers while EEGs of the remaining subsets (Sets C, D and E) were collected intra-cranially from five epilepsy patients. Sets A and B consisted of segments taken from surface EEG recordings that were carried out using a standardized 10–20 electrode placement scheme. Set A was recorded in an awakened state with eyes open and Set B was taken in the same state with eyes closed. Specifically, Set C was collected from the hippocampal formation, located in the opposite hemisphere of epileptogenic zone; whereas Sets D and E were measured from the epileptogenic zone. Sets C/D and E were collected when the epilepsy patients were in inter-ictal and ictal phases, respectively. Since Sets A,B and Sets C,D,E are collected from different sources, Sets C and D are blended together to constitute the inter-ictal class while Set E is used to form the ictal class, and our objective is to classify inter-ictal and ictal EEGs from single channel EEG recordings accurately. All the five sets of EEG data were cut out from continuous multichannel EEG recordings after visual inspection for artifacts due to muscle activity or eye movement. The recording was done with the same 128-channel amplifier system, using an average common reference. The data were sampled at 173 samples per second and digitized using 12 bit resolution.

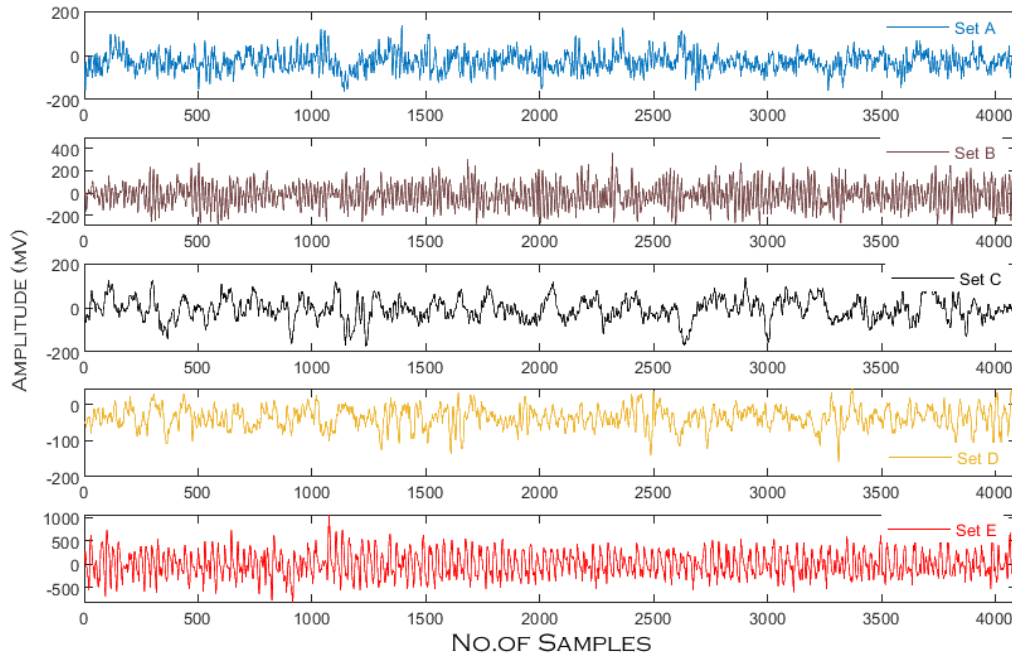


Figure 7

The above figure represents the diagrammatic plot representation of sample EEG's of ictal and inter-ictal classes.

3.2 Maximum Overlap Discrete Wavelet Transform MODWT:

The method of extracting features from the time domain has been proven efficient/effective by using wavelet analysis over that of many other approaches (Eg., Fourier transforms),due to the main adverse effect being the exclusion of primary features while analysis.{Rajee and Shahabi,2016}.The wavelet transform (WT)has been categorized into discrete(DWT) and continuous(CWT) ,and both are capable of decomposing time domain signals into separate sub-signals

The following flowchart gives an overall procedural explanation of the DWT wavelet decomposition upon each level.

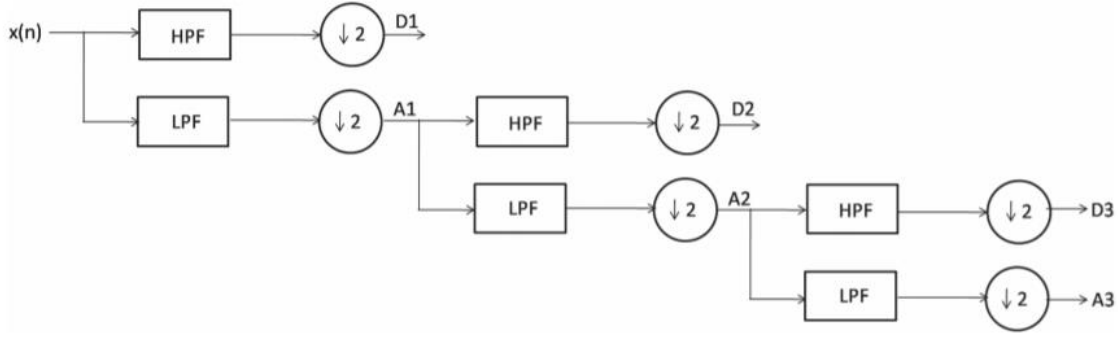


Figure 8

The number of levels is truly independent upon the sample database and the selection of classes which is to be classified.

In the above flow chart the output coefficients sum up to $N+1$ coefficients, where N being the no of levels predetermined. The variables D_i and A_i , where $i=1,2,3,\dots$ indicates the Detailed coefficients and Approximate coefficients obtained as outputs from High pass filter and Low pass filters respectively.

However DWT includes short comings such as,

- (i) DWT requires the length of time-series data to be of a dyadic nature, the power of 2 $\{2^j\}$ j denotes the level of wavelet decomposition, and
- (ii) A moderate shift in its origin can influence a wavelet data's output performance, leading to the issue of circular shift [28]. Consequently, it may be difficult to align the transformed TS (time series/domain) signal over the passage of time, practically causing a loss of information during a model's training phase and therefore, influence the model's result [29].

Therefore, an updated method has been undertaken where the sub-sampling process is avoided, leading to a higher level of efficiency in data, in the resulting wavelet and scaling coefficients on comparison with DWT.

MODWT is shift-invariant and can be applied to the time series (domain) of independent sample size .

With this algorithm, the signal is easy to interpret [31] and provides a non-orthonormal approach applicable to any size of the modelled TS-based data [29,31]. The area of concern lies in computational efficiency of MODWT which is lesser than DWT and to retain the energy preserving property of DWT scaling filters are rescale along with the MODWT wavelets.

The procedural criteria of application of MODWT on the database is similar to that of DWT except few dissimilarities upon the resultant obtained , that being MODWT generates twice the number of results as DWT hence it is highly redundant.

MODWT can eliminate ‘alignment’ artifacts, but its redundancies are problematic for some uses.

- unlike the DWT, the MODWT is – defined for all sample sizes (no ‘power of 2’ restrictions)
- unaffected by circular shifts to X in that coefficients, details and smooths shift along with X
- highly redundant in that a level J_0 transform consists of $(J_0 + 1)N$ values rather than just N
- unlike the DWT, MODWT is not orthonormal (in fact MODWT is highly redundant)
- unlike the DWT, MODWT is defined naturally for all samples sizes (i.e., N need not be a multiple of a power of two)
- unlike the DWT, MODWT discrete wavelet power spectrum same for X and its circular shifts $T^m X$

Similar to the DWT, the MODWT offers,

- multi resolution analyses (MRAs) , but with certain additional desirable features; e.g., unlike the DWT, MODWT-based MRA has details and smooths that shift along with X .
- similar to the DWT, an analysis of variance (ANOVA) can be based on MODWT wavelet coefficients
- a scale-based multi resolution analysis
- a scale-based analysis of the sample variance
- a pyramid algorithm for computing the transform efficiently

Properties of MODWT:

- as was true with the DWT, we can use the MODWT to obtain – a scale-based additive decomposition (MRA) and – a scale-based energy decomposition (ANOVA)
- in addition, the MODWT can be computed efficiently via a pyramid algorithm

MODWT ALGORITHM :

- DWT and MODWT have different normalizations for filters, and there is no downsampling by 2^j in the MODWT

The MODWT determines scaling and wavelet coefficients by applying low and high pass filters to the original dataset. i.e, if wavelet coefficients ($d_{j,l}$) and scaling coefficients ($s_{j,l}$) are used in DWT, MODWT will consist of wavelet ($\check{d}_{j,l}$) and scaling filters ($\check{s}_{j,l}$) as follow (Mouatadid et al., 2019)

where,

j =decomposition level and $j=1,2,\dots$:

l =filter length and $l=1,2,\dots$

$$\check{d}_{j,l} = d_{j,l} / 2^{j/2}$$

$$\check{s}_{j,l} = s_{j,l} / 2^{j/2}$$

The MODWT wavelet and scaling coefficients for an original time series dataset (e.g. $\{Y_t, t = 0, 1, \dots, N-1\}$) can then be computed by the following equations as of (Zhu et al., 2014)

$$\check{W}_{j,l} = \sum_{l=0}^{j-1} \check{d}_{j,l} Y_{t-l \bmod N}$$

The above equation is addressed to calculate the wavelet coefficients ($\check{W}_{j,l}$) or the respective original time series database. Similarly the scaling coefficients ($\check{V}_{j,l}$) can be processed using the following equation,

$$\check{V}_{j,l} = \sum_{l=0}^{Lj-1} \check{s}_{j,l} Y_{t-l \bmod N}$$

In the above equation ,

$$L_j = (2^j - 1)(L - 1) + 1$$

3.3 FEATURE EXTRACTION TECHNIQUE

3.3.1 LOCAL BINARY PATTERN

LBP (Local Binary Pattern) is a feature extraction technique that is used to extract best features of a given signal. Here, An LBP code for a neighborhood of sampled data is produced by thresholding the neighboring samples against center samples of a processing

window. This procedure is iteratively done across the entire signal and a segment of the 1-D signal is alternatively described by a sparser occurrence histogram of LBP codes. The 1-D LBP operator is adapted from the 2-D LBP. It examines a neighborhood of data samples from a signal and assigns an LBP code to each centre sample after thresholding them against the neighboring samples.

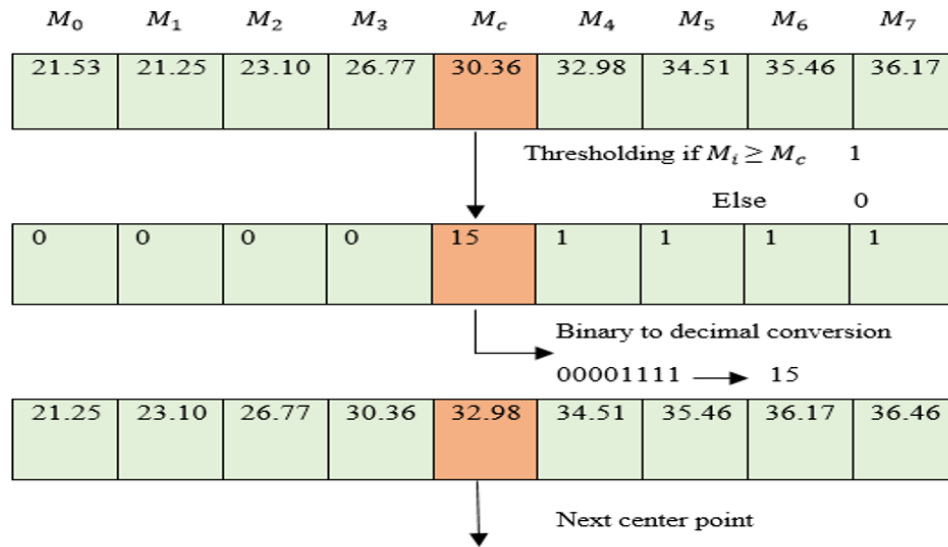


Figure 9

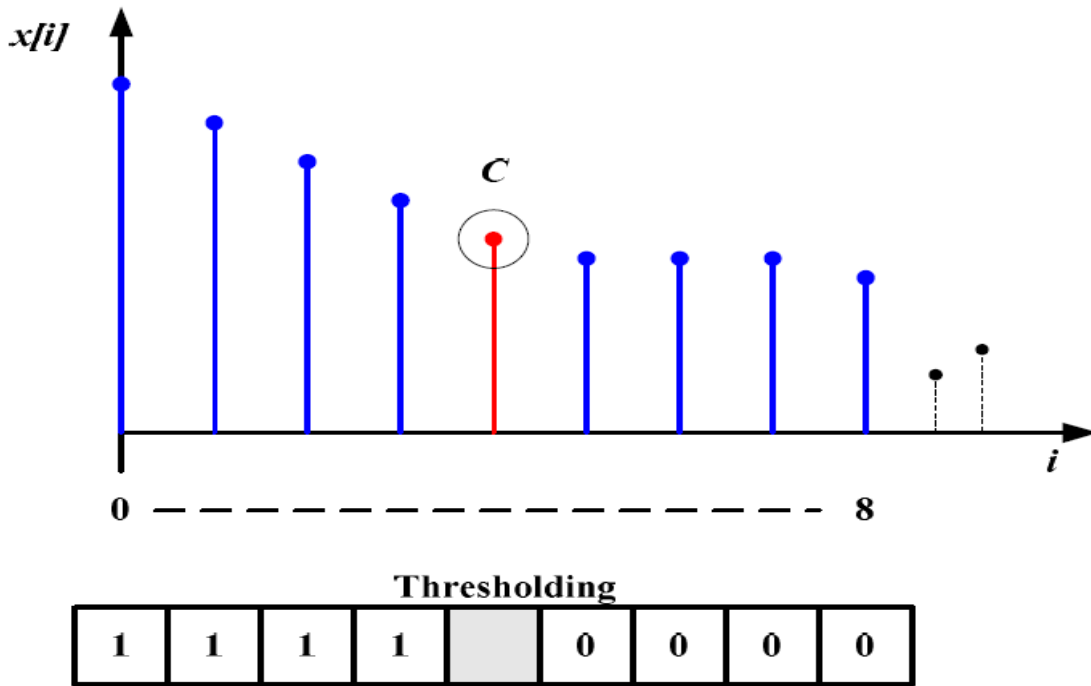


Figure 10

An illustration of the 1-D LBP operator is given in the Figure10 where P is set to 8 and the centre sample C is circled. The 8 neighboring samples are thresholded against C to produce a binary code of 1111_0000. This code is then multiplied by the binomial weights given to the corresponding samples and the obtained values are summed to give the resulting LBP code of 15. The LBP codes can locally describe the data using the difference between a sample and its neighbors. For a constant or slowly varying signal, these differences cluster near zero. At peaks and troughs, the difference will be relatively large, whereas at edges, the differences in some directions will be larger than those from other directions. This method of feature extraction technique has proven to be more effective when compared to other pattern and feature extraction techniques. A uniform pattern in LBP is defined by an LBP code which has at most two one-to-zero or zero-to-one transitions. The remaining non-uniform patterns are assigned to a single histogram

bin and each uniform pattern is assigned to a separate bin.

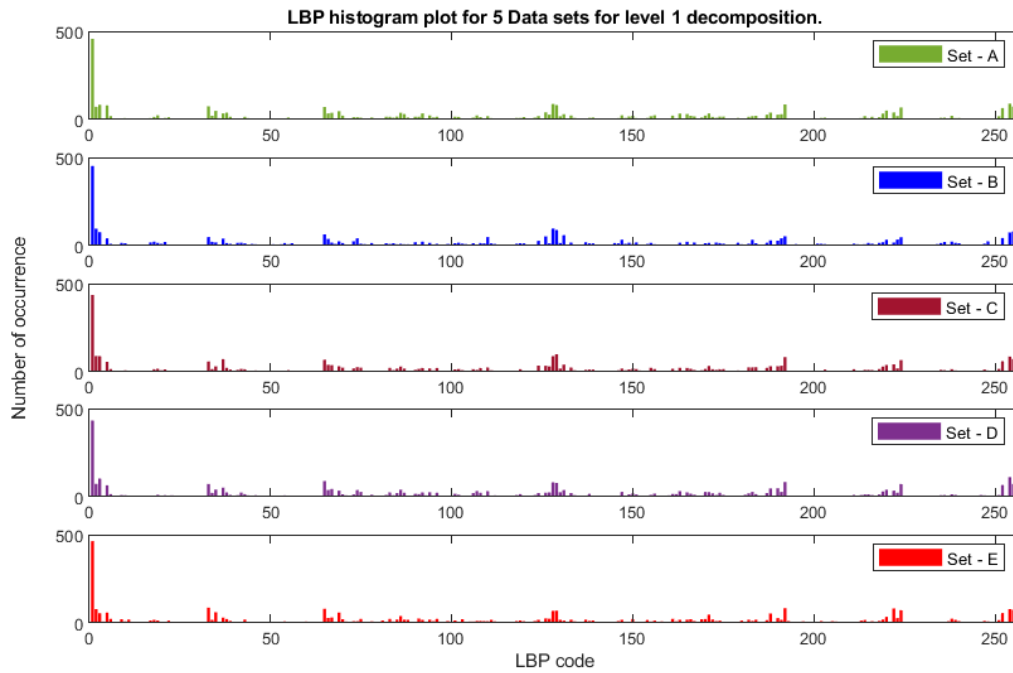


Figure 11

3.3.2 LOCAL GRADIENT PATTERN

Though LBP has been adopted by researchers for quite a period, the long histogram created by Local Binary Pattern is not suitable for a large-scale facial database. Under some certain circumstances, it misses the local feature, as it does not consider the effect of the centre pixel. LGP technique preserves the structural property of a pattern. In order to obtain maximum accuracy in the proposed technique, the classifier has to be trained with the best features as the input. The local gradient pattern technique has been employed to extract the best features possible from the dataset. These features are fed into the classifier for training.

Local gradient pattern is a local descriptor technique that constitute the database samples in order of nine corresponding cells that hold up each sample signals respectively. Once the centre sample signal is selected, the gradient value is calculated for the neighbouring

samples. The resultant sample signal set's mean is calculated using the mathematical formula,

$$g_{avg} = \frac{1}{m} \sum_{i=0}^{m-1} g_i$$

The mean value is then reduced from the neighbouring samples existing in the cells. Post reduction from the neighbouring samples the if the respective result is positive ($g_{ci} \geq 0$) the resultant is denoted using 1 else 0. The decimal denomination of the binary digits is then calculated for the centre sample signal. Similarly, the neighbouring samples are assigned to act as the center cell value and thus the LGP representative value of the corresponding sample is obtained. Finally a histogram representation of the obtained LGP like sample signal resultants is accumulated accordingly for the classifier input.

LGP Mechanism:

- Denote the samples in cells of nine in count
- Elect the centre sample
- Find gradient of the neighbouring samples
- Calculate the mean
- Subtract the mean value from each neighbouring samples
- If ($g_{ci} \geq 0$) resultant value is 1, else 0.
- Calculate the decimal equivalent
- Create histogram for like values

The mathematical representation of the whole LGP algorithm is ,

$$1D-LGP_m(S_c) = \sum_{i=0}^{m-1} s(|P_i - S_c| - \frac{1}{m} \sum_{i=0}^{m-1} (|P_i - S_c|) 2^i$$

Where,

$$s(x) = \begin{cases} 1, & \text{if } x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

The pictorial representation of local binary pattern algorithm is as follows,

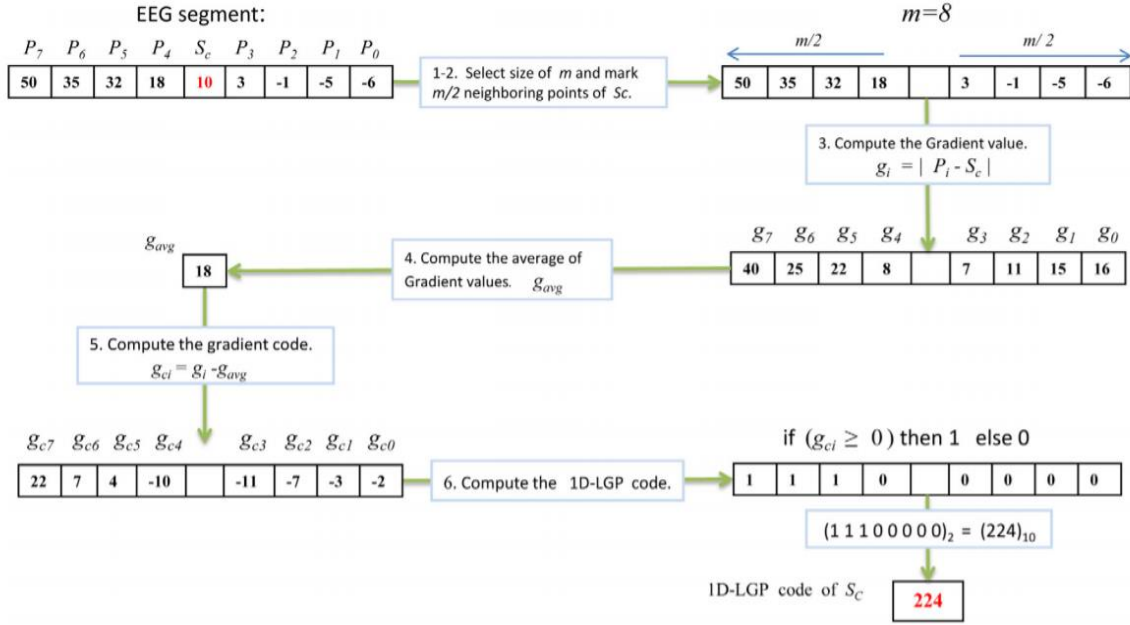


Figure 11

The variables P_i and S_c are associated to neighbouring cells and centre sample respectively where as g_{ci} is the gradient code computed after $g_{ci} = g_i - g_{avg}$.

CHAPTER 4

CLASSIFIER

4.1 ARTIFICIAL NEURAL NETWORK

Artificial neural networks (ANN) or connectionist systems are computing systems vaguely inspired by the biological neural networks that constitute animal brains.^[1] Such systems "learn" to perform tasks by considering examples, generally without being programmed with task-specific rules. For example, in image recognition, they might learn to identify images that contain cats by analyzing example images that have been manually labeled as "cat" or "no cat" and using the results to identify cats in other images. They do this without any prior knowledge of cats, for example, that they have fur, tails, whiskers and cat-like faces. Instead, they automatically generate identifying characteristics from the examples that they process.

An ANN is based on a collection of connected units or nodes called artificial neurons, which loosely model the neurons in a biological brain. Each connection, like the synapses in a biological brain, can transmit a signal to other neurons. An artificial neuron that receives a signal then processes it and can signal neurons connected to it.

In ANN implementations, the "signal" at a connection is a real number, and the output of each neuron is computed by some non-linear function of the sum of its inputs. The connections are called edges. Neurons and edges typically have a weight that adjusts as learning proceeds. The weight increases or decreases the strength of the signal at a connection. Neurons may have a threshold such that a signal is sent only if the aggregate signal crosses that threshold. Typically, neurons are aggregated into layers. Different layers may perform different transformations on their inputs. Signals travel from the first layer (the input layer), to the last layer (the output layer), possibly after traversing the layers multiple times.

The original goal of the ANN approach was to solve problems in the same way that a human brain would. But over time, attention moved to performing specific tasks, leading to deviations from biology. ANNs have been used on a variety of tasks, including computer vision, speech recognition, machine translation, social network filtering, playing board and video games, medical diagnosis, and even in activities that have traditionally been considered as reserved to humans, like painting.

As a classifier, artificial neural network (ANN) is notable due to its powerful capabilities. For this reason, behaviors and performances of neural network training algorithms were investigated and compared on classification task of the EEG signals in this study.

Training algorithms of neural network were categorized in five group as Gradient Descent, Resilient Backpropagation, Conjugate Gradient, Quasi-Newton, and Levenberg-Marquardt. Conjugate Gradient Backpropagation was performed during the training and test stages to achieve more generalized results. Furthermore, several evaluation parameters, such as accuracy (ACC), sensitivity (Se), specificity (Sp), and geometric mean (GM), were taken into account during performance comparison of the algorithms.

The Train Neural Network was done using `traincgb` train function proposed by Powell-Beale.

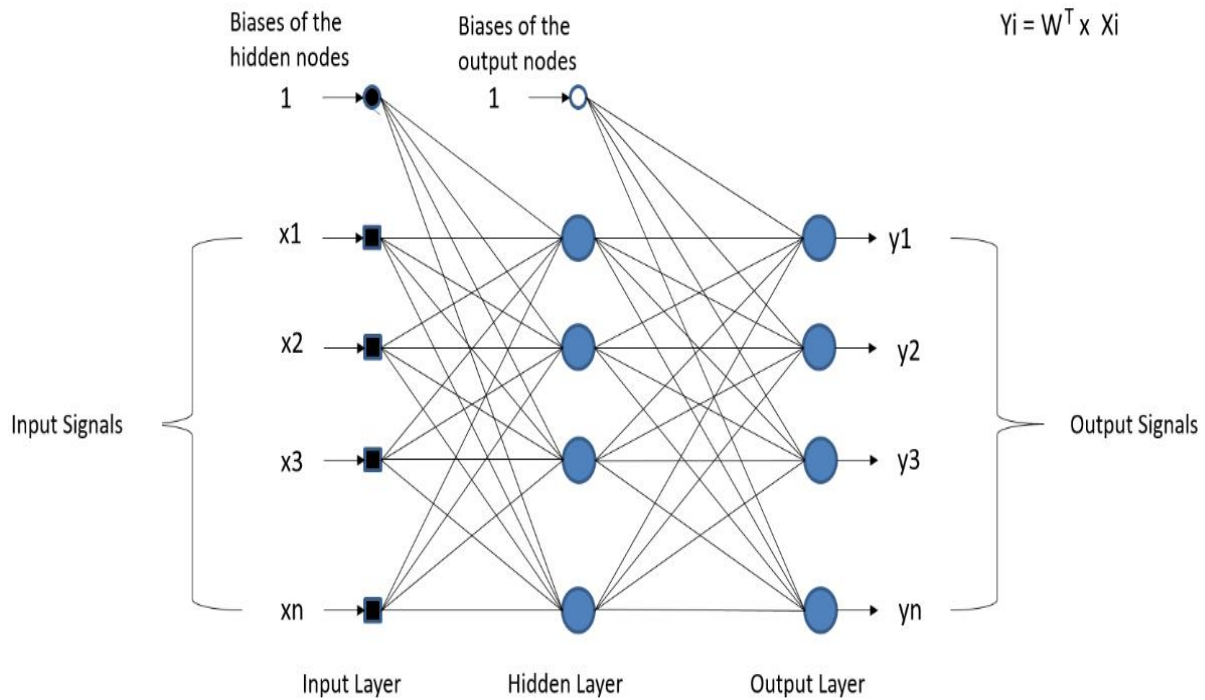


Figure 12

4.2 CONJUGATE GRADIENT ALGORITHMS

The basic backpropagation algorithm adjusts the weights in the steepest descent direction (negative of the gradient). This is the direction in which the performance function is decreasing most rapidly. It turns out that, although the function decreases most rapidly along the negative of the gradient, this does not necessarily produce the fastest convergence. In the conjugate gradient algorithms a search is performed along conjugate directions, which produces generally faster convergence than steepest descent directions. In this section, we present four different variations of conjugate gradient algorithms.

In most of the training algorithms that we discussed up to this point, a learning rate is used to determine the length of the weight update (step size). In most of the conjugate gradient algorithms, the step size is adjusted at each iteration. A search is made along the

conjugate gradient direction to determine the step size, which minimizes the performance function along that line. There are five different search functions included in the toolbox, and these are discussed at the end of this section. Any of these search functions can be used interchangeably with a variety of the training functions described in the remainder of this chapter. Some search functions are best suited to certain training functions, although the optimum choice can vary according to the specific application. An appropriate default search function is assigned to each training function, but this can be modified by the user.

4.3 POWELL - BEALE ALGORITHM

For all conjugate gradient algorithms, the search direction is periodically reset to the negative of the gradient. The standard reset point occurs when the number of iterations is equal to the number of network parameters (weights and biases), but there are other reset methods that can improve the efficiency of training. One such reset method was proposed by Powell, based on an earlier version proposed by Beale. This technique restarts if there is very little orthogonality left between the current gradient and the previous gradient.

The `traincgb` routine has somewhat better performance than `traincgp` for some problems, although performance on any given problem is difficult to predict. The storage requirements for the Powell-Beale algorithm (six vectors) are slightly larger than for Polak-Ribière (four vectors).

Training stops when any of these conditions occurs:

- The maximum number of epochs (repetitions) is reached.
- The maximum amount of time is exceeded.
- Performance is minimized to the goal.
- The performance gradient falls below `min_grad`.

- Validation performance has increased more than max_fail times since the last time it decreased (when using validation).

4.4 KRUSKAL - WALLIS TEST

Kruskal–Wallis test was adopted to select the significant statistical and entropy features. The Kruskal-Wallis test is used to examine the significance of the features. Non-significant features are discarded. The Kruskal-Wallis test calculates the probability of being wrong when concluding that there is no difference between three or more treatment groups. The treatments commonly affect the individual response levels in the treatment groups. There is therefore an order among the responses to the treatments where one treatment tends to give the lowest responses, another the next lowest responses and so forth. It can be mathematically represented as:

$$T = \frac{12}{N(N+1)} \sum_{i=1}^k \frac{R_i^2}{N_i} - 3 (N+1)$$

Where,

N= total number of observations

R_i = the rank for an individual sample

K= the number of groups

N_i = the number of observations in group i

In order to compare our results with the results provided by previous works on these datasets, we used the experimental procedure: a ten-fold crossvalidation approach. Cross-validation is a statistical method used to estimate the skill of machine learning models.

It is commonly used in applied machine learning to compare and select a model for a given predictive modeling problem because it is easy to understand, easy to implement, and results in skill estimates that generally have a lower bias than other methods.

Cross-validation is a resampling procedure used to evaluate machine learning models on a limited data sample.

The procedure has a single parameter called k that refers to the number of groups that a given data sample is to be split into. As such, the procedure is often called k -fold cross-validation. When a specific value for k is chosen, it may be used in place of k in the reference to the model, such as $k=10$ becoming 10-fold cross-validation.

Cross-validation is primarily used in applied machine learning to estimate the skill of a machine learning model on unseen data. That is, to use a limited sample in order to estimate how the model is expected to perform in general when used to make predictions on data not used during the training of the model.

It is a popular method because it is simple to understand and because it generally results in a less biased or less optimistic estimate of the model skill than other methods, such as a simple train/test split.

The general procedure is as follows:

1. Shuffle the dataset randomly.
2. Split the dataset into k groups
3. For each unique group:
 - Take the group as a hold out or test data set
 - Take the remaining groups as a training data set
 - Fit a model on the training set and evaluate it on the test set
 - Retain the evaluation score and discard the model

4. Summarize the skill of the model using the sample of model evaluation scores

Importantly, each observation in the data sample is assigned to an individual group and stays in that group for the duration of the procedure. This means that each sample is given the opportunity to be used in the hold out set 1 time and used to train the model $k-1$ times.

Three common tactics for choosing a value for k are as follows:

- Representative: The value for k is chosen such that each train/test group of data samples is large enough to be statistically representative of the broader dataset.
- $k=10$: The value for k is fixed to 10, a value that has been found through experimentation to generally result in a model skill estimate with low bias a modest variance.
- $k=n$: The value for k is fixed to n , where n is the size of the dataset to give each test sample an opportunity to be used in the hold out dataset. This approach is called leave-one-out cross-validation.

There are a number of variations on the k -fold cross validation procedure. Three commonly used variations are as follows:

- Train/Test Split: Taken to one extreme, k may be set to 2 (not 1) such that a single train/test split is created to evaluate the model.
- LOOCV: Taken to another extreme, k may be set to the total number of observations in the dataset such that each observation is given a chance to be the held out of the dataset. This is called leave-one-out cross-validation, or LOOCV for short.
- Stratified: The splitting of data into folds may be governed by criteria such as ensuring that each fold has the same proportion of observations with a given categorical value, such as the class outcome value. This is called stratified cross-validation.
- Repeated: This is where the k -fold cross-validation procedure is repeated n times, where importantly, the data sample is shuffled prior to each repetition, which results in a different split of the sample.

CHAPTER 5

RESULTS AND DISCUSSIONS

The simulation for load forecasting using BONN dataset is done using MATLAB18a (R2018a). MATLAB® is a high-level language and interactive environment that enables us to perform computationally intensive tasks faster than with other traditional programming languages.

5.1 RESULTS

The Bonn dataset is transformed using MODWT. The feature extraction from the transformed signal is done using two different feature extraction methods namely LBP (local binary pattern) and LGP (local gradient pattern). The levels of extraction are $n+1$ i.e., $6+1 = 7$. So the number of features per co-efficient is 256. So total number of feature is $7 \times 256 = 1792$. So for each input signal we get 1792 features. These features are further optimised using Kruskal-Wallis test.

The following calculations are done at the output:

- (i) Accuracy
- (ii) Sensitivity
- (iii) Specificity
- (iv) Positive Predictive Value (PPV)
- (v) Negative Predictive Value (NPV)
- (vi) F-1 Score
- (vii) G-Mean

S.NO	PERFORMANCE METRICS	FORMULA	DESCRIPTION
1	Accuracy	$\frac{TP + TN}{TP + FP + TN + FN}$	It is the ratio of correct decisions made by a classifier
2	Sensitivity	$\frac{TP}{TP + FN}$	It measures how much a classifier can recognize positive samples
3	Specificity	$\frac{TN}{FP + FN}$	It measures how much a classifier can recognize negative samples
4	Positive Predictive Value (PPV)	$\frac{TP}{TP + FP}$	It is the proportion of positive results in statistics and diagnostic tests
5	Negative Predictive Value (NPV)	$\frac{TN}{FN + TN}$	It is the proportion of negative results in statistics and diagnostic tests
6	Matthews correlation coefficient (MCC)	$\frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$	It is a correlation coefficient between the observed and predicted classifications
7	F-1 Score	$2 * \frac{\text{precision} * \text{recall}}{\text{precision} + \text{recall}}$	harmonic mean of precision and recall
8	G-mean	$\sqrt{\text{Sensitivity} * \text{Specificity}}$	measures the balance between classification performances on both the classes of data

Table 1 shows the mathematical calculations done at the output.

db8	Feature Extraction techniques							
	LBP				LGP			
	AE	ABCD_E	A_D_E	A_C_E	AE	ABCD_E	A_D_E	A_C_E
Accuracy	99.55	97.34	94	94.533	99.35	97.4	93.4333	93.733
Sensitivity	0.1581	0.1687	0.82450	1.51975	0.06324	0.16841	1.3841	NaN
Specificity	0.12649	1.58194	1.05334	1.3380	0.3179	1.55856	1.10921	NaN
PPV	0.14784	0.35602	NaN	NaN	0.2757	0.34508	NaN	NaN
NPV	0.12162	NaN	NaN	NaN	0.07027	NaN	NaN	NaN
MCC	0.2771	NaN	NaN	NaN	0.36179	NaN	NaN	NaN
F1-Score	0.16032	0.23882	NaN	NaN	0.17660	0.20466	NaN	NaN
G-Mean	0.14868	1.07188	1.31541	2.23716	0.19599	1.16658	2.1666	NaN

Table 2 shows the output values obtained from features extracted through LBP and LGP using db8 wavelet.

db12	Feature Extraction techniques							
	LBP				LGP			
	AE	ABCD_E	A_D_E	A_C_E	AE	ABCD_E	A_D_E	A_C_E
Accuracy	98.75	97.3	93.533	95.1666	98.95	96.62	90.633	94.266
Sensitivity	0.22135	0.1521	0.8216	1.1408	0.12649	0.16203	1.9892	1.421
Specificity	0.51524	1.5247	1.50767	0.6298	0.50561	2.1537	2.2435	1.10003
PPV	0.43782	0.3326	NaN	NaN	0.4500	0.45438	NaN	NaN
NPV	0.2797	NaN	NaN	NaN	0.14054	NaN	NaN	NaN
MCC	0.7058	NaN	NaN	NaN	0.61006	NaN	NaN	NaN
F1-Score	0.32708	0.1959	NaN	NaN	0.29759	0.27042	NaN	NaN
G-Mean	0.36875	1.16724	1.63344	1.09567	0.32250	1.8462	3.3072	1.7473

Table 3 shows the output values obtained from features extracted through LBP and LGP using db12 wavelet.

db16	Feature Extraction techniques							
	LBP				LGP			
	AE	ABCD_E	A_D_E	A_C_E	AE	ABCD_E	A_D_E	A_C_E
Accuracy	99.05	97.88	92.6	93.4333	99	97.88	91.133	94.7
Sensitivity	0.06324	0.12231	1.13213	1.84084	0.15811	0.1022	2.6146	1.3482
Specificity	0.39668	1.30901	1.30901	1.45188	0.4220	1.6280	1.4755	1.28175
PPV	0.35684	0.29713	0.24313	0.28515	0.38822	0.35009	NaN	NaN
NPV	0.05749	0.05445	0.05855	0.05321	0.16290	NaN	NaN	NaN
MCC	0.39275	0.32455	0.36312	0.36688	0.55634	NaN	NaN	NaN
F1-Score	0.19768	0.17823	0.18433	0.18857	0.27580	0.20466	NaN	NaN
G-Mean	0.21287	1.17317	1.84707	2.36528	0.28932	1.33906	2.8766	1.8297

Table 4 shows the output values obtained from features extracted through LBP and LGP using db16 wavelet.

db20	Feature Extraction techniques							
	LBP				LGP			
	AE	ABCD_E	A_D_E	A_C_E	AE	ABCD_E	A_D_E	A_C_E
Accuracy	98.75	97.84	93.133	92.733	98.85	97.7	92.366	92.5
Sensitivity	0.1581	0.1304	1.3681	1.4338	0.2529	0.1589	1.50923	1.54675
Specificity	0.5163	1.7225	1.1703	1.4330	0.39001	1.6829	2.25917	1.8849
PPV	0.4475	0.3666	NaN	NaN	0.39142	0.3692	NaN	NaN
NPV	0.1897	NaN	NaN	NaN	0.2180	NaN	NaN	NaN
MCC	0.6465	NaN	NaN	NaN	0.61862	NaN	NaN	NaN
F1-Score	0.3082	0.21954	NaN	NaN	0.32818	0.2378	NaN	NaN
G-Mean	0.34109	1.42997	1.9324	2.2058	0.3240	1.28765	2.9489	2.72543

Table 5 shows the output values obtained from features extracted through LBP and LGP using db20 wavelet.

sym8	Feature Extraction techniques							
	LBP				LGP			
	AE	ABCD_E	A_D_E	A_C_E	AE	ABCD_E	A_D_E	A_C_E
Accuracy	99.25	96.52	94.833	94.5	99.25	96.52	94.833	94.5
Sensitivity	0.1054	0.16754	1.2144	1.4036	0.1054	0.16754	1.2144	1.4036
Specificity	0.3204	2.0339	1.1805	1.0512	0.3204	2.03392	1.1805	1.05129
PPV	0.2504	0.4445	NaN	NaN	0.25048	0.44454	NaN	NaN
NPV	0.1297	NaN	NaN	NaN	0.1297	NaN	NaN	NaN
MCC	0.3926	NaN	NaN	NaN	0.39264	NaN	NaN	NaN
F1-Score	0.18174	0.2767	NaN	NaN	0.18174	0.2767	NaN	NaN
G-Mean	0.22170	1.55472	2.0809	1.4603	0.22170	1.5547	2.0809	1.4603

Table 6 shows the output values obtained from features extracted through LBP and LGP using sym8 wavelet.

sym 12	Feature Extraction techniques							
	LBP				LGP			
	AE	ABCD_E	A_D_E	A_C_E	AE	ABCD_E	A_D_E	A_C_E
Accuracy	99.9	97.68	90.433	94.333	99.9	97.4	93.4333	93.733
Sensitivity	0	0.1260	2.2958	1.6242	0.06324	0.16841	1.3841	NaN
Specificity	0.06324	1.3514	1.5001	1.2510	0.3179	1.55856	1.10921	NaN
PPV	0.05749	0.29984	NaN	NaN	0.2757	0.34508	NaN	NaN
NPV	0	NaN	NaN	NaN	0.07027	NaN	NaN	NaN
MCC	0.06037	NaN	NaN	NaN	0.36179	NaN	NaN	NaN
F1-Score	0.03011	0.18712	NaN	NaN	0.17660	0.20466	NaN	NaN
G-Mean	0.03245	0.97871	2.92657	1.87136	0.19599	1.16658	2.1666	NaN

Table 7 shows the output values obtained from features extracted through LBP and LGP using sym12 wavelet.

sym16	Feature Extraction techniques							
	LBP				LGP			
	AE	ABCD_E	A_D_E	A_C_E	AE	ABCD_E	A_D_E	A_C_E
Accuracy	96.95	96.58	92.8	93.433	96.95	96.58	92.8	93.4333
Sensitivity	0.88543	0.1668	0.7653	1.3107	0.88543	0.16688	0.76535	1.3107
Specificity	0.71963	2.05438	1.7969	1.9947	0.71963	2.0543	1.7969	1.9947
PPV	NaN	0.43846	NaN	NaN	NaN	0.43846	NaN	NaN
NPV	0.61247	NaN	NaN	NaN	0.61247	NaN	NaN	NaN
MCC	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
F1-Score	NaN	0.2637	NaN	NaN	NaN	0.26375	NaN	NaN
G-Mean	0.96871	1.8535	2.19124	2.38723	0.96871	1.8535	2.19124	2.3872

Table 8 shows the output values obtained from features extracted through LBP and LGP using sym16 wavelet.

sym20	Feature Extraction techniques							
	LBP				LGP			
	AE	ABCD_E	A_D_E	A_C_E	AE	ABCD_E	A_D_E	A_C_E
Accuracy	98.75	97.54	90.8	93.733	98.75	97.54	90.8	93.733
Sensitivity	0.18973	0.0928	1.9372	1.2829	0.18973	0.09280	1.9372	1.2829
Specificity	0.46644	1.7612	1.6096	1.0720	0.4664	1.7612	1.60964	1.07206
PPV	0.41655	0.3697	NaN	NaN	0.4165	0.36971	NaN	NaN
NPV	0.18446	NaN	NaN	NaN	0.1844	NaN	NaN	NaN
MCC	0.6047	NaN	NaN	NaN	0.6047	NaN	NaN	NaN
F1-Score	0.30484	0.2195	NaN	NaN	0.30484	0.21953	NaN	NaN
G-Mean	0.3218	1.54914	2.64569	1.5080	0.32180	1.54914	2.6456	1.50808

Table 9 shows the output values obtained from features extracted through LBP and LGP using sym20 wavelet.

PAPERS	A-E (Healthy vs Ictal)	ABCD-E (Non-ictal vs Ictal)	A-D-E (Healthy vs Non focal inter-ictal vs Ictal)	A-C-E (Healthy vs focal inter-ictal vs Ictal)
THIS PAPER	99.9%(LBP-sym12)	97.88%	94.83%	95.16%
Rubén San-Segundo et.al	99.8%	99.5%	96.5%	95.7%
Anubha Gpta et.al	94.85%	97.79%	Nil	Nil
Y.Kumar et.al	98%	92.4%	Nil	91.5%
Erdem Yavuz et.al	99.5%	98.74%	97.33%	Nil
Arpit Bhardwaj et.al	85.647%	80%	Nil	Nil

Table 10 shows the comparison of the obtained output values with the available papers proposed by other authors.

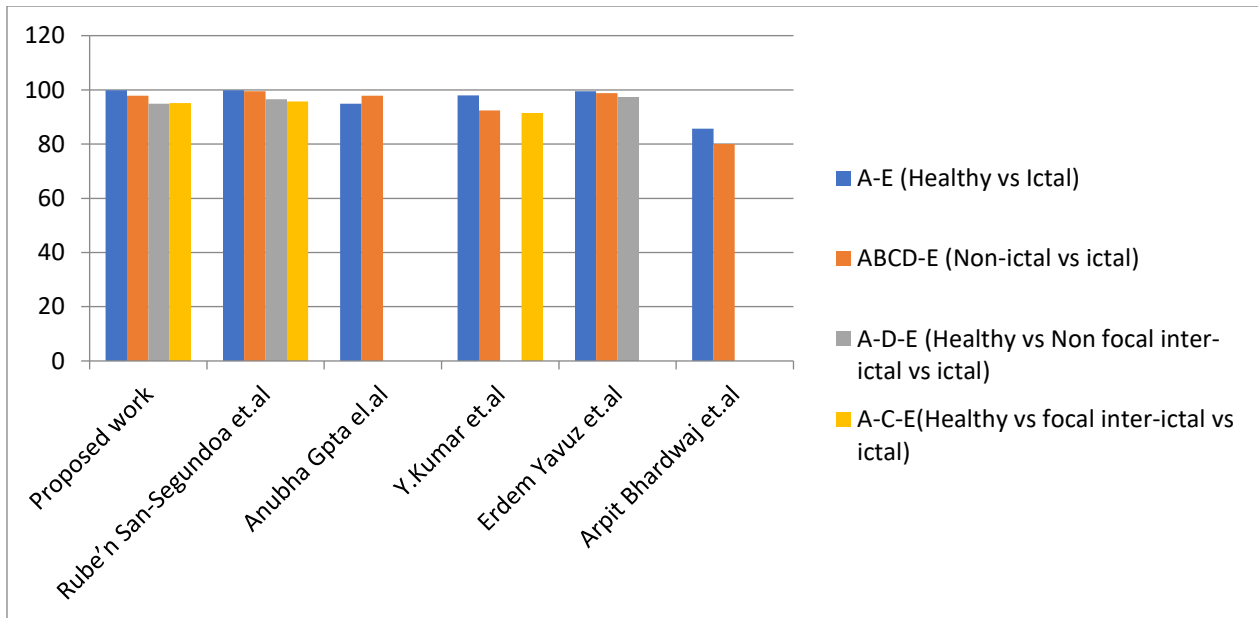


Figure 13

Project	Cases	Accuracy %
M. Peker et.al	A-E	100
Y.Kumar et.al	A-E	100
S. Xie et.al	A-E	100
Z. Iscan et.al	A-E	100
Proposed work	A-E	99.9
L. Guo et.al	A-E	99.85
N.S. Tawfik et.al	A-E	99.8
D.Wang et.al	A-E	99.5
K.Polat et.al	A-E	99.1
V.P. Nigam et.al	A-E	98.7
H.Ocak et.al	A-E	97.5
L. Guo et.al	A-E	96
A.Subasi et.al	A-E	95.2
N.Kannath al et.al	A-E	94.5

Table 11 shows the comparison of A-E case with various works done in the past.

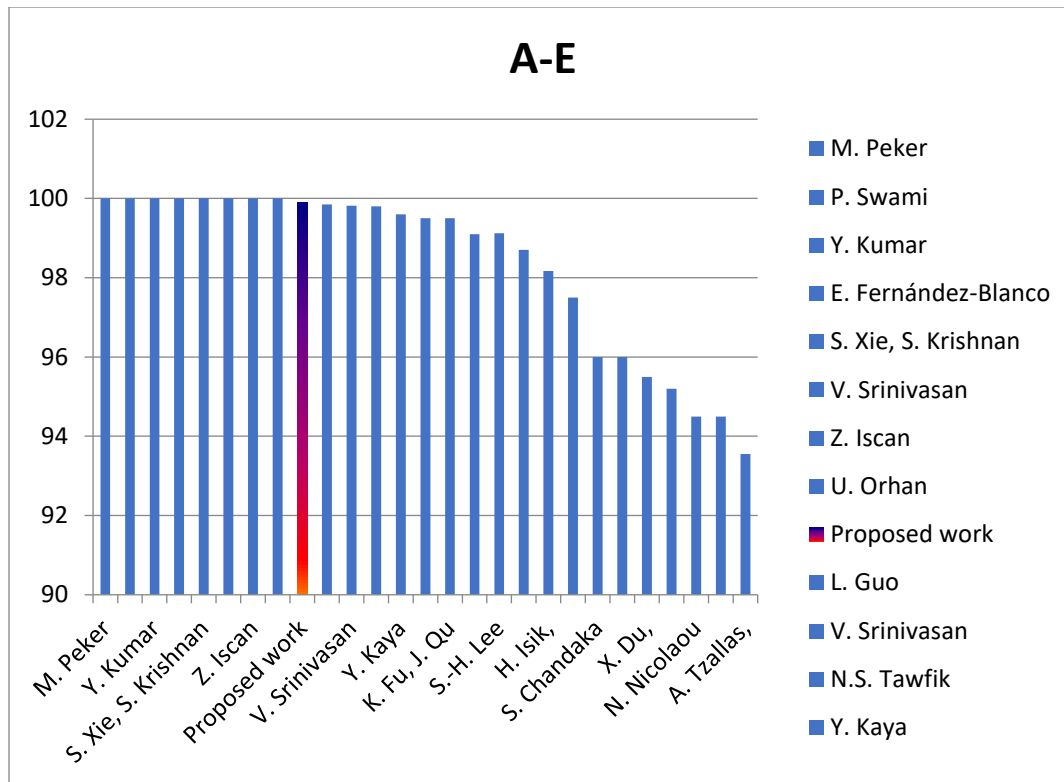


Figure 14

Figure 14 shows the graphical representation of the A-E case comparison

Project	Cases	Accuracy %
S.Xie et.al	ABCD-E	100
E.Fernandez-Blanco et.al	ABCD-E	100
M.Peker et.al	ABCD-E	99.15
L.Guo et.al	ABCD-E	98.72
Proposed work	ABCD-E	97.88
Y. Kumar et.al	ABCD-E	98.65

Table 12 shows the comparison of ABCD-E case with various works done in the past

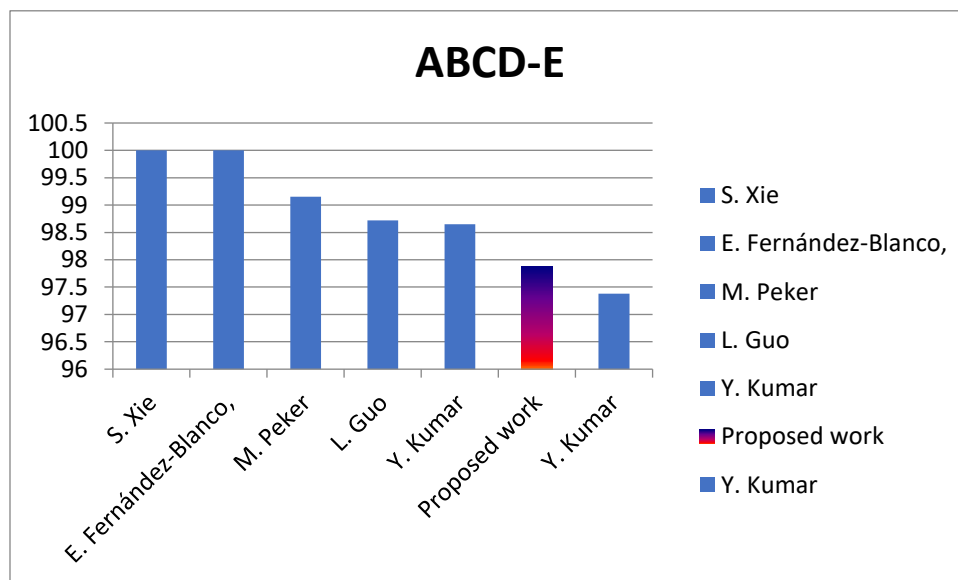


Figure 14

Figure 14 shows the graphical representation of the A-E case comparison

Project	Cases	Accuracy %
M.Peker	A-D-E	99.3
E.Fernandez-Blanco et.al	A-D-E	99.25
U.Orhan	A-D-E	98.22
Y.Kaya et.al	A-D-E	97.06
Proposed work	A-D-E	94.83
Y. Kumar et.al	A-D-E	98.65

Table 13 shows the comparison of A-D-E case with various works done in the past

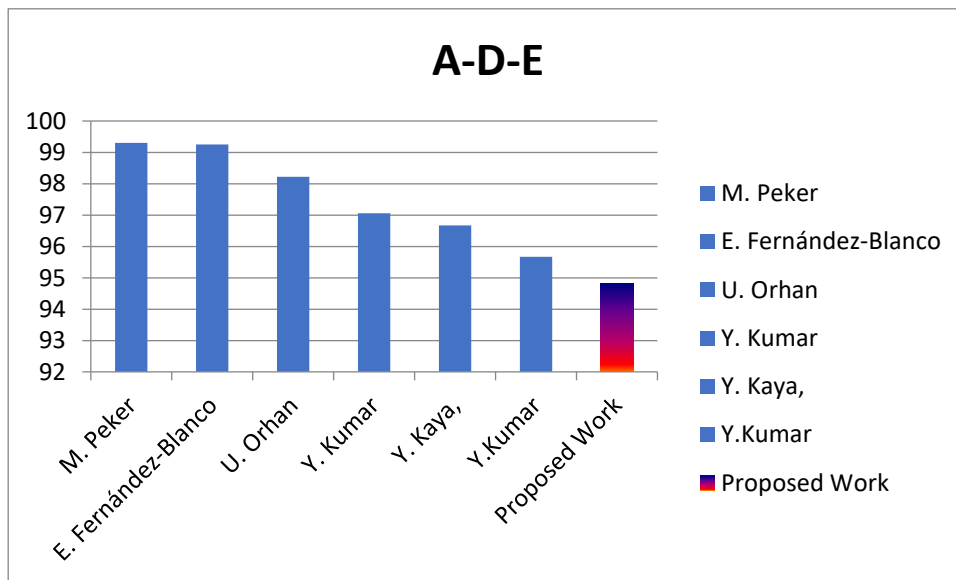


Figure 15

Figure 15 shows the graphical representation of the A-D-E case comparison

5.2 CONCLUSION

Wavelet-based analysis approaches have shown tremendous potential in distinguishing diverse types of EEGs. MODWT has undeniable advantages over DWT which makes it the best way to use for signal transformation. The fusion of various feature extraction techniques and the ANN algorithm makes the work more optimised to get more efficient output. The feature extraction techniques such as LBP and LGP extracts $7 \times 256 = 1792$ features which is a large number of collection to choose the best features from, by the Kruscal-Wallis test method, which again is the best way possible for feature selection. The Conjugate Gradient Backpropagation classifier algorithm has proven effective throughout the history in signal classification. And the k-fold crossvalidation method makes the training and testing of the learning classifier algorithm more efficient in the prediction of the weight factors which in turn brings out the possibility of higher accuracy and other output features. This is very evident because of the accuracy 99.9% in the normal vs ictal (A-E) case while using LBP feature extraction technique and the sym-20 wavelet based analysis. This proves the domination of the fusion techniques over the other ways of operation.

CHAPTER 8

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